Communication Protocols

System Design Crash Course: Quick Revision Before Interviews

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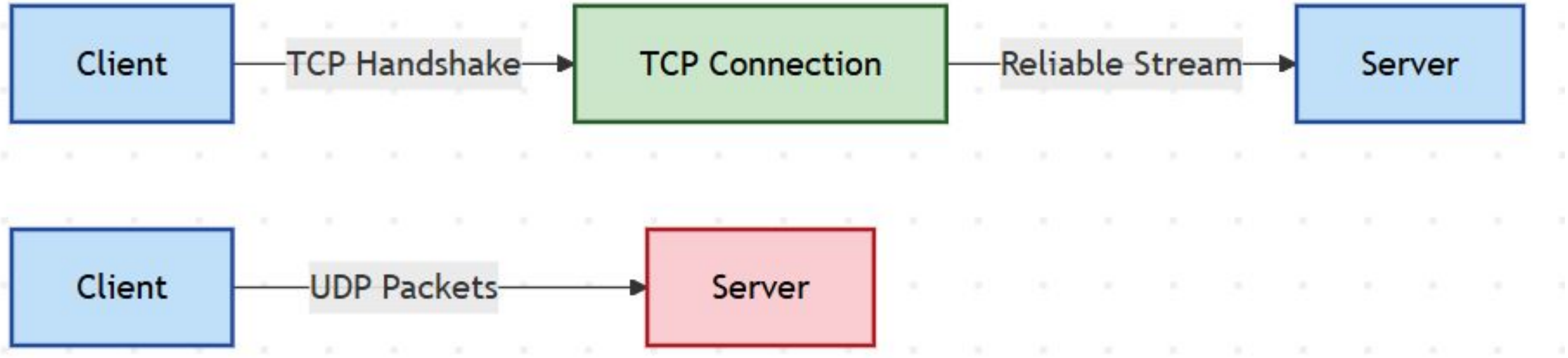
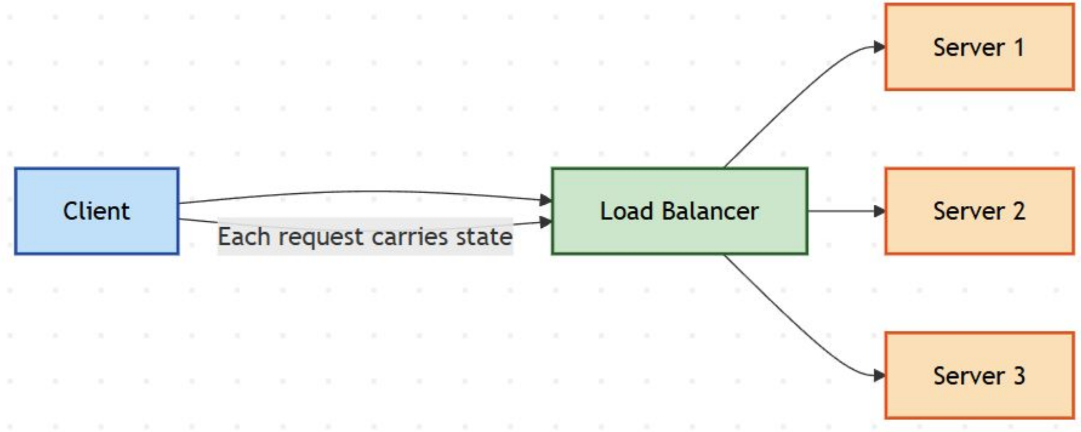
TCP vs UDP

- TCP: reliable, ordered delivery
- UDP: fast, connectionless delivery
- TCP handles retries and congestion
- UDP favors latency over guarantees
- Reliability versus speed trade-off



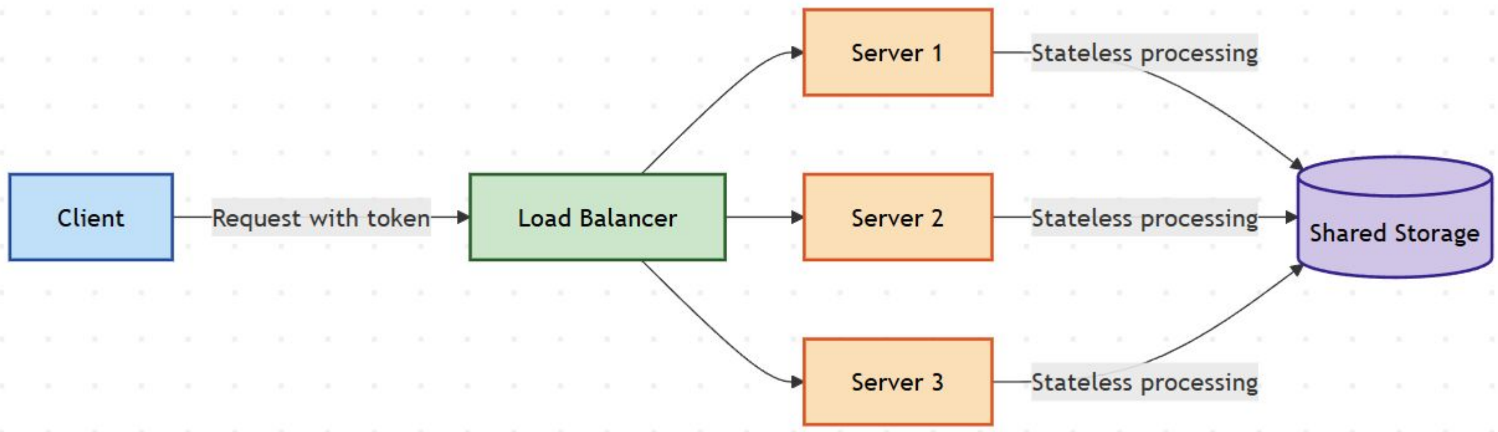
Why HTTP Scales So Well

- Stateless request–response model
- No server-side session dependency
- Easy horizontal scaling
- Works well with load balancers
- State pushed to clients



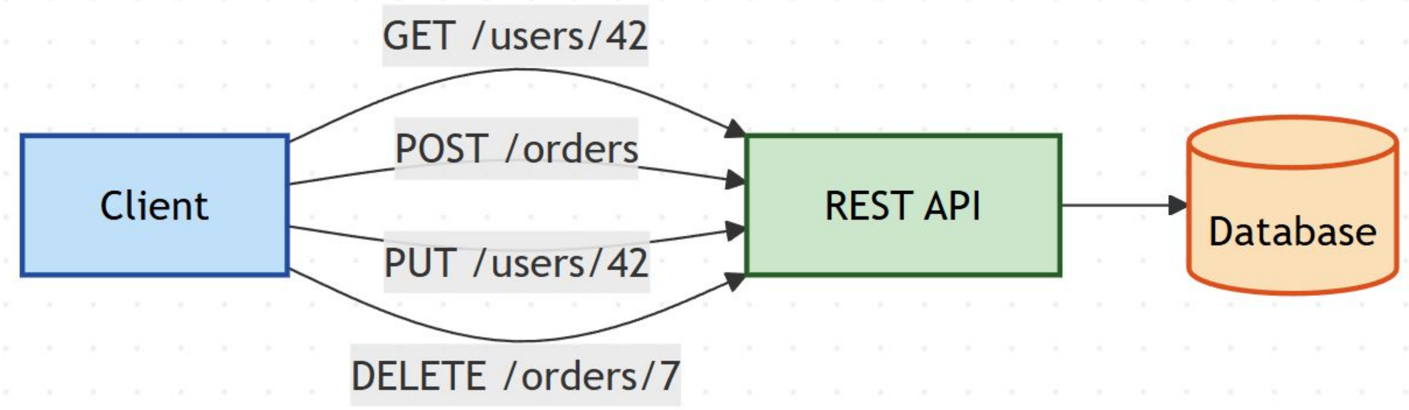
HTTP Stateless Nature

- Servers store no client state
- Each request is independent
- Enables horizontal scaling
- Simplifies failure handling
- State moved to clients or storage



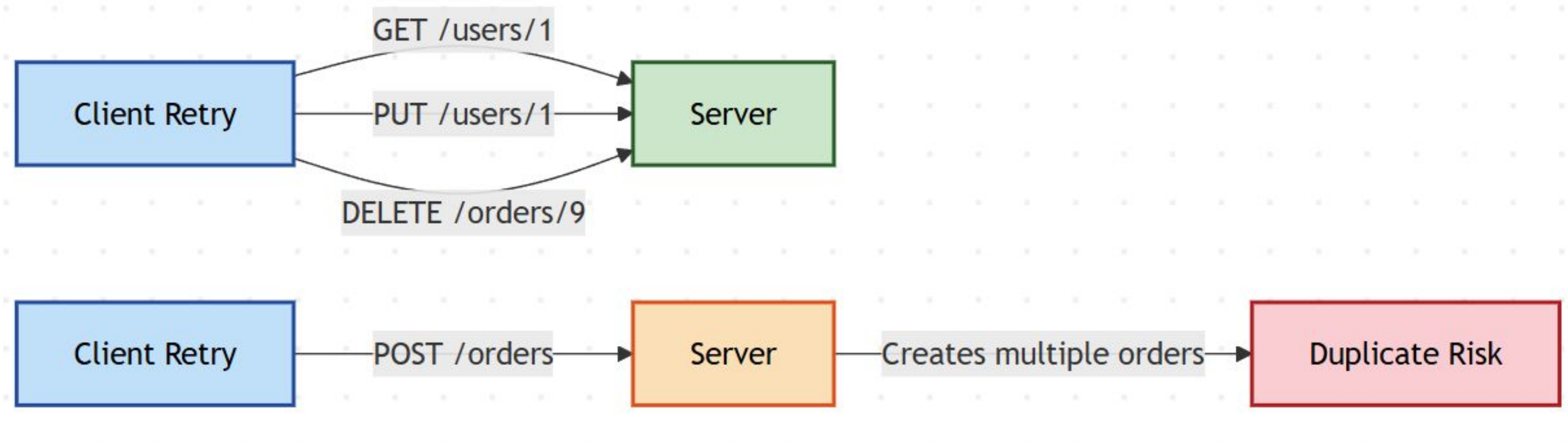
REST – Interview Perspective

- Resource-oriented API design
- Uses standard HTTP verbs
- Stateless interactions
- Predictable URLs and responses
- Simplicity over strict purity



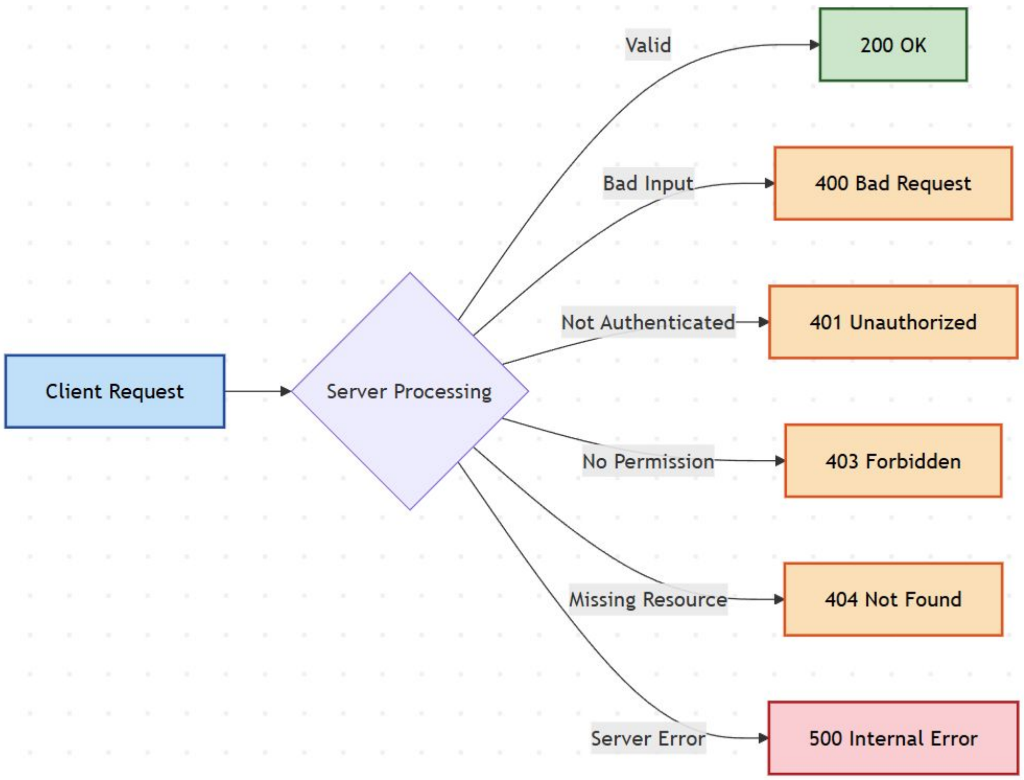
HTTP Methods and Idempotency

- GET: read-only, idempotent
- POST: create, not idempotent
- PUT: replace, idempotent
- DELETE: remove, idempotent
- Idempotency enables safe retries



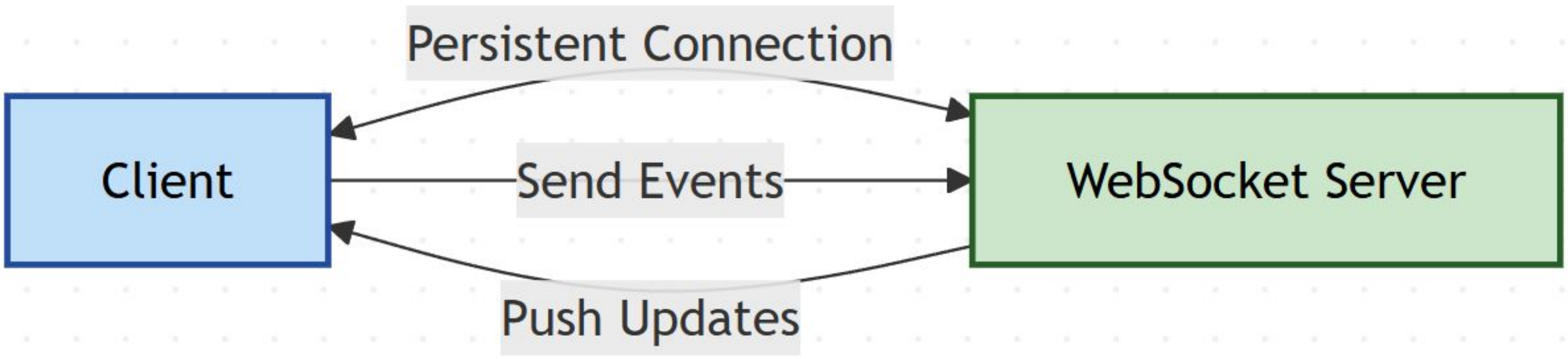
HTTP Status Codes

- 200: successful request
- 400: bad client request
- 401 vs 403: auth vs permission
- 404: resource not found
- 500: server-side failure



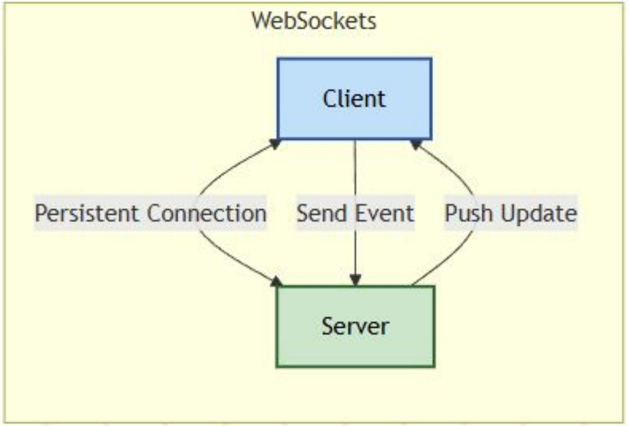
WebSockets

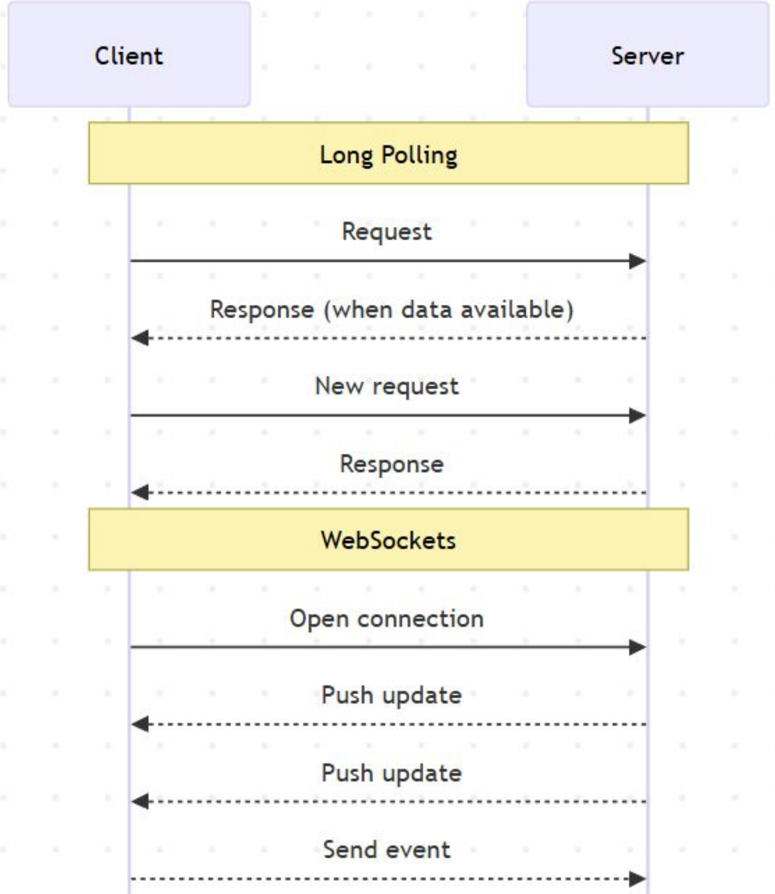
- Persistent client–server connection
- Full-duplex, bidirectional communication
- Server can push data
- Low-latency real-time updates
- Higher resource and state cost



Long Polling vs WebSockets

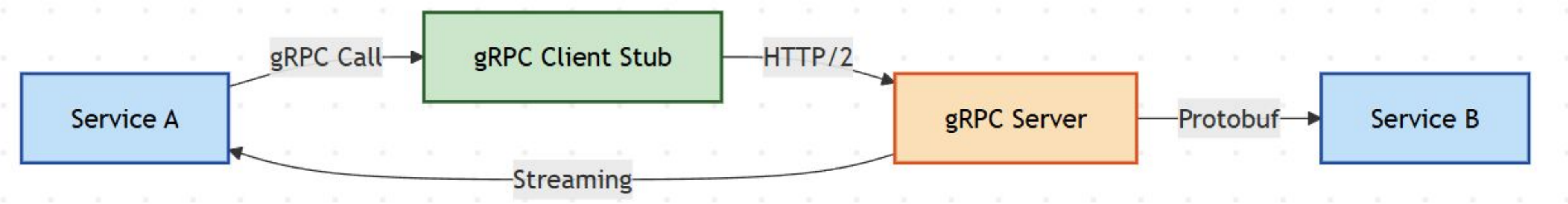
- Long polling uses repeated HTTP requests
- WebSockets use persistent connection
- Long polling wastes network resources
- WebSockets enable real-time push
- Simplicity versus efficiency trade-off





gRPC

- RPC framework over HTTP/2
- Uses Protocol Buffers
- Strongly typed contracts
- Supports streaming
- Optimized for internal services



GraphQL

- Client defines response shape
- Avoids over- and under-fetching
- Single endpoint, multiple queries
- Strong schema and typing
- More server-side complexity